

# STA 332/MATH 343: Midterm

Duke University, Spring 2026

**Instructions:** You have 75 minutes to complete the following questions. This exam is closed book and no electronic devices are allowed. You must justify all assumptions, clearly cite any theorems used, and *show all your work*.

Your name: \_\_\_\_\_

(Turn over when exam starts)

**Problem 1** [5+5+5=15 points]

Provide a brief (but formal) definition of the following concepts:

(a) Minimal sufficient statistic.

(b) Complete statistic.

(c) UMVUE.

**Problem 2** [15 points]

Let  $T = (T_1, T_2)$  be a sufficient statistic for  $\theta$  in a model  $\{f_\theta\}$ . Which of the following statistics are we *sure* to be sufficient for  $\theta$ ? (Briefly justify each answer.)

$$S_1 = (T_1 + T_2, T_1 - T_2), \quad S_2 = (T_1^2, T_2), \quad S_3 = (T_1, T_2 \cdot \theta), \quad S_4 = (T_1, e^{2T_1 + T_2})$$

**Problem 3** [6+6+8=20 points]

Let  $X_i \stackrel{\text{iid}}{\sim} f_\theta(x)$  for  $i = 1, \dots, n$ , where

$$f_\theta(x) = \frac{1}{\theta} \mathbf{1}_{\{\theta < x < 2\theta\}}, \quad \theta > 0.$$

(a) Prove that  $\theta$  is a *scale* parameter.

(b) Find a minimal sufficient statistic for  $\theta$ . Justify your answer.

(c) Is it complete? Justify your answer.

**Problem 4** [50 points + 10 additional points]

Let  $X_i \stackrel{\text{iid}}{\sim} f_\theta(x)$  for  $i = 1, \dots, n$ , where

$$f_\theta(x) = \theta x^{\theta-1}, \quad 0 < x < 1, \theta > 0.$$

(a) Determine the distribution of  $Y = -\log(X)$  for  $X \sim f_\theta(x)$ .

(b) Find a sufficient and complete statistic  $T$  for  $\theta$ .

(c) Compute  $E[T]$ .

(d) Find the UMVUE for  $1/\theta$ .

- (e) Find the MLE for  $1/\theta$ .
- (f) (Bonus 10 points) Determine the distribution of  $T$ .



